

Supplementary materials for: A Multi-Stream Temporal Context Model for narrative memory

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S0. Architectural lineage of MS-TCM

The present paper describes MS-TCM as a multi-stream extension of Cornell and Zhang^{S1} hierarchical CMR. Two architectural revisions preceded the current state. The original v1 formulation (notes/ms-tcm.pdf) used a single global context plus a bank of storyline-specific context vectors mixed into an encoding composite with fixed weights, plus a task-dependent retrieval reweighting; that architecture failed to reproduce the canonical free-recall phenomena (serial position curve, probability of first recall, lag-CRP) because it bypassed the standard CMR retrieval route, and was retired in favour of the v6 design (notes/two_level_cmr_v6.pdf) which rebases MS-TCM on top of Cornell and Zhang^{S1}. The v6 spec proposed a single new mechanism over the C&Z baseline: storyline-return reinstatement at encoding with strength λ (v6 Eq. 6).

In implementing v6, two further additions emerged from the FRFR-category worked example. First, the C&Z 2025 baseline structurally cannot produce a primacy spike in the probability of first recall: the recency-driven initiation rule places its mass at the end of the list even when participants frequently begin recall from $sp = 1$ items. This motivated the τ *storyline-initiation mixture* (retrieval section of the main text). Second, the late-list serial position scallop and within-category clustering in random presentation lists motivated a second iteration with strict hierarchical retrieval and a separate global cross-storyline context (Iteration 2 in the main text). The full iteration-by-iteration log of empirical motivations and architectural decisions is at notes/mstcm_architecture_log.md. The v1 $\rightarrow$ v6 symbol inventory and file-level deletion manifest is at notes/v6_migration.md. Git history preserves the v1 source; this supplement and the main text use v6 notation augmented with the τ and Iteration 2 additions.

S1. Off-by-one correction in the reformatted recall table

The upstream exp2.egg file from^{S3} encodes each recall event's provenance in a temporal HDF5 attribute that is 0-indexed against the $i\{k\}$ subgroup names in the presented group. The initial reformat (scripts/reformat_frfr_category.py, commit 8684fa0) treated temporal as if it were 1-indexed, which shifted every in-list recall's serial position down by one and silently re-coded the

last word of every list as an extra-list intrusion. The corrected mapping (commit 405e272, `sp = temporal + 1` for `temporal ∈ [0, 15]`, else `sp = 0`) restores the expected serial-position distribution (primacy at SP1–2, recency at SP14–16) and reduces the extra-list intrusion count from 399 to 10.

A regression test at `code/tests/test_frfr.py::test_recall_serial_position_shape` asserts $P(\text{recall}) > 0.3$ for the last three serial positions and an intrusion rate below 5%, so any future off-by-one would fail CI.

S2. Likelihood-surface flatness on small synthetic data

At FRFR-scale real data the MS-TCM likelihood surface is informative (grid scans across the active parameters $(\beta_{\text{enc}}, \gamma_{\text{ic}})$ give a range of ~ 100 nats across the parameter simplex, vs a uniform-recall baseline ~ 1200 nats below the MLE). At the smaller scales used for quick sanity checks, the same grid scan shows an essentially flat surface (LL varies by ~ 2 nats around a uniform baseline). The practical consequence is that parameter-recovery tests need a synthetic dataset at FRFR scale ($30 \text{ pt} \times 16 \text{ lists} \times 16 \text{ words} \times 4 \text{ categories} \approx 7,680$ recalls) before the L-BFGS-B optimizer can reliably recover the true parameter vector; smaller datasets converge to whatever corner the initial point leans toward. This is a property of the likelihood surface, not a bug in the fitter.

S3. Implementation details: norm-preserving drift, softmax gain, primacy, no-repeats

Four implementation choices beyond the minimal equations in the main text are necessary for the model’s forward simulator and its posterior-predictive to match the observed behavioural regularities (SPC, P(first recall), lag-CRP, clustering). Each is standard in the CMR / hierarchical-CMR literature but is worth stating explicitly.

Unit-norm feature encoder. Equation (1) of the main text assumes unit-norm inputs so the drift invariant $\|\mathbf{c}(t)\| = 1$ is preserved at every step. Raw multi-hot feature vectors (with k active slots) have norm $\sqrt{k}$; the feature encoder therefore L_2 -normalises each row by default. See `code/ms_tcm/features.py (encode_features(..., normalize=True))`.

Norm-preserving ρ under non-orthogonal inputs. For strictly orthonormal inputs, Howard and Kahana^{S2}’s $\rho = \sqrt{1 - \beta^2}$ preserves $\|\mathbf{c}(t)\| = 1$ exactly. Multi-hot feature vectors that share active slots across items are unit-norm but not orthogonal, so the orthonormal formula no longer maintains the invariant. We adopt the generalised formula $\rho = \sqrt{1 + \beta^2(\hat{c}^2 - 1)} - \beta \hat{c}$ (where $\hat{c} = \mathbf{c}_{\text{prev}} \cdot \mathbf{c}^N$) from Polyn et al.^{S4} Eq. 3, which reduces exactly to $\sqrt{1 - \beta^2}$ when $\hat{c} = 0$ and preserves unit norm for arbitrary unit-norm inputs. This is applied at both the item level (rate β_{enc}) and the storyline level (rate β_{story}) in the v6 hierarchical layout; see `code/ms_tcm/drift.py::norm_preserving_rho`.

Softmax gain k . The retrieval readout $a_j = M^{\text{IC}^\top} \mathbf{c}^{\text{ret}}$ produces activations whose range is bounded; the softmax gain $k > 0$ sharpens the distribution. Without k , the softmax $P(j) \propto \exp(a_j)$ over bounded activations is close to uniform and the model fails to produce recency or contiguity

signatures. k is fit alongside the other parameters; it is initialized at the Cornell and Zhang (2025) Table 1 value $k = 6.50$.

Primacy gradient. Standard CMR inherits a Polyn-et-al.-style primacy scaling $p(t) = 1 + \phi \exp(-\psi t)$ applied to the M^{IC} update magnitude at encoding step t . This is a v6-native CMR mechanism, not an MS-TCM extension. Under the `--standard-tcm` reduction the hierarchical layer is disabled and the implementation falls back to a stronger primacy gradient ($\phi = 30.0$, $\psi = 0.8$) to keep the shape assertion satisfied; under MS-TCM the hierarchical layer contributes additional primacy through the list-start synchronization of the storyline-level context, so a modest gradient ($\phi = 1.5$, $\psi = 0.5$) is sufficient.

No-repeats candidate set. Participants almost never repeat an item within one recall sequence (intra-list repeat rates below 1%). A softmax readout without an exclusion constraint, however, would predict the most recent item with near-certainty whenever k is moderate. We therefore mask the softmax output to zero for any serial position already recalled on the current list before re-normalising — the standard CMR convention. The likelihood routine skips observed repeats (rather than returning $-\infty$) so the handful of real repeats in the data do not poison the dataset log-likelihood.

S4. Numerical-accuracy anchors and model comparison

The v1 MS-TCM numerical anchor (composite-similarity closed form at $\beta = 0.5$, $m = 3$) is retired alongside the v1 architecture: the present model does not form an encoding composite and has no analog of the v1 mixture weights. Two anchors take its place. First, the Cornell and Zhang^{S1} hierarchical free-recall implementation in `code/ms_tcm/likelihood_core.py` is verified to 10^{-10} tolerance against hand-derived oracle tests at `code/tests/test_hcmr_oracle.py`, so the comparator against which MS-TCM’s ΔLL is reported is itself a faithful implementation of its published inheritance. Second, when MS-TCM is configured with $\lambda = 0$, $\tau = 0$, and a single storyline ($K = 1$), it reduces exactly to that C&Z baseline at the same machine-epsilon tolerance — a regression covered by `code/tests/test_hcmr_standard_tcm.py`. The Layer 1 shape assertions of `code/tests/test_behavioral_regression.py` pin the canonical free-recall signatures (SPC, pFR, lag-CRP shapes) for both the C&Z baseline and MS-TCM Iteration 1. Together these anchors establish that any reported ΔLL between MS-TCM and the C&Z baseline is attributable to the new mechanisms (λ , τ , per-storyline contexts) rather than to implementation drift.

Supplementary references

- [S1] Charlotte A. Cornell and Qiong Zhang. Hierarchical context guides human memory search. *Psychological Review*, 2025. Preprint in `notes/CornZhan25.pdf`; DOI to be assigned upon publication.
- [S2] M W Howard and M J Kahana. A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46:269–299, 2002.
- [S3] Jeremy R Manning, Emily C Whitaker, Paxton C Fitzpatrick, Madeline R Lee, Allison M Frantz, Bryan J Bollinger, Darya Romanova, Campbell E Field, and Andrew C Heusser. Feature and order manipulations in a free recall task affect memory for current and future lists, 2023. PsyArXiv preprint, <https://psyarxiv.com/erzfp>. Data and code: <https://github.com/ContextLab/FRFR-analyses>.
- [S4] S M Polyn, K A Norman, and M J Kahana. Task context and organization in free recall. *Neuropsychologia*, 47:2158–2163, 2009.